

Sato-Tate Groups and Distributions of

$$y^2 = x^{p^2} - 1$$

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The Sato-Tate Conjecture (Cont.)

Let C be a smooth, projective, genus g curve over $\mathbb{Q}$.

- Originally posed when A is an elliptic curve ($g = 1$), can be extended to higher-genus curves via $\text{Jac}(C)$.

Denote the normalized L-polynomial of primes p of good reduction for C as

$$\bar{L}_p(C, T) = T^{2g} + a_1 T^{2g-1} + a_2 T^{2g-2} + \dots + a_2 T^2 + a_1 T + 1.$$

As $p \rightarrow \infty$, we can realize distributions of $\bar{L}_p(C, T)$'s coefficients as [moment sequences](#).

Note: For each prime $p \nmid \ell$ of good reduction, $\text{Frob}_p \in \text{Gal}(\bar{F}/F)$ is mapped to a conjugacy class under $\rho_{A, \ell}$ in $\text{ST}(\text{Jac}(C_{p^2}))$. The conjecture is equivalent to talking about limiting distributions of Frobenius elements' conjugacy classes

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(Generalized) Sato-Tate Conjecture

As $p \rightarrow \infty$, the distribution of coefficients of $\bar{L}_p(C, T)$ converges to the distributions of $\text{ST}(\text{Jac}(C))$'s conjugacy classes' charpoly coefficients via the Haar measure.

Note: For each prime $p \nmid \ell$ of good reduction, $\text{Frob}_p \in \text{Gal}(\bar{F}/F)$ is mapped to a conjugacy class under $\rho_{A, \ell}$ in $\text{ST}(\text{Jac}(C_{p^2}))$. The conjecture is equivalent to talking about limiting distributions of Frobenius elements' conjugacy classes

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We want to see what $ST(\text{Jac}(C_{p^2}))$ and its distributions look like.

Computing Sato-Tate Groups

We need to compute two objects:

$$ST^0(\mathrm{Jac}(C_{p^2})) \quad \text{and} \quad ST(\mathrm{Jac}(C_{p^2}))/ST^0(\mathrm{Jac}(C_{p^2})).$$

Computing $ST(\text{Jac}(C_{p^2}))/ST^0(\text{Jac}(C_{p^2}))$

First, we have that the endomorphism field of $\text{Jac}(C_{p^2})$ is $\mathbb{Q}(\zeta_{p^2})$ ([GGL24, Prop. 3.5.1]). By [GGL25, Thm. 7.2.12], this is also its *connected monodromy field*. So,

$$ST(\text{Jac}(C_{p^2}))/ST^0(\text{Jac}(C_{p^2})) \cong \text{Gal}(\mathbb{Q}(\zeta_{p^2})/\mathbb{Q}).$$

Moreover

$$\text{Gal}(\mathbb{Q}(\zeta_{p^2})/\mathbb{Q}) \cong (\mathbb{Z}/p^2\mathbb{Z})^\times,$$

so $ST(\text{Jac}(C_{p^2}))/ST^0(\text{Jac}(C_{p^2}))$ is cyclic (because $(\mathbb{Z}/p^2\mathbb{Z})^\times$ is) and has order $\phi(p^2)$.

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To find a generator of $ST(\text{Jac}(C_{p^2}))/ST^0(\text{Jac}(C_{p^2}))$, we study *endomorphisms* of $\text{Jac}(C_{p^2})$ acted on by $\text{Gal}(\mathbb{Q}(\zeta_{p^2})/\mathbb{Q})$.

Computing $ST(\text{Jac}(C_{p^2}))/ST^0(\text{Jac}(C_{p^2}))$ (Cont.)

Let $Z := -\text{diag}(\zeta_{p^2}, \bar{\zeta}_{p^2})$. Endomorphisms of $\text{Jac}(C_{p^2})$ are of the form

$$\alpha = \text{diag}(Z, Z^2, Z^3, \dots, Z^g),$$

where $g = (p^2 - 1)/2$ is the genus of C_{p^2} .

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By computing a $\langle \sigma_a \rangle = \text{Gal}(\mathbb{Q}(\zeta_{p^2})/\mathbb{Q})$ (through Sage), the action of σ_a on α ($\sigma_a Z^t = Z^{at}$) **either only permutes or permutes and conjugates entries of α** . **Tracking this behavior gives the component group generator.**

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Let I be the 2×2 identity matrix,

$$J := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and $\langle n \rangle_{p^2}$ denote $n \pmod{p^2}$.

Computing $ST(\text{Jac}(C_{p^2}))/ST^0(\text{Jac}(C_{p^2}))$ (Cont.)

Proposition [CGHM25]

The $2g \times 2g$ matrix γ (in $USp(2g)$) defined by

$$\gamma[i, j] = \begin{cases} 1 & \text{if } j = \langle ai \rangle_{p^2} \\ J & \text{if } j = p^2 - \langle ai \rangle_{p^2} \\ 0 & \text{otherwise.} \end{cases}$$

generates the component group of $ST(\text{Jac}(C_{p^2}))$.

Proof idea:

- Show that $\gamma\alpha\gamma^{-1} = \sigma_a\alpha$ (shows that $\gamma \in ST(\text{Jac}(C_{p^2}))$)
- $|\gamma| = \phi(p^2)$ (order is equal to that of the component group).

Example of a Component Group Generator: C_{25}

When $p = 5$, using σ_2 as a generator for $\text{Gal}(\mathbb{Q}(\zeta_{25})/\mathbb{Q})$ (found via Sage) gives

$$\gamma = \begin{pmatrix} 0 & I & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & I \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & J & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & J & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & J & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & J & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & J & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ J & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Here, $g = (25 - 1)/2 = 12$. So, γ is a 24×24 matrix. For $p = 7$, γ is a 48×48 matrix!

The Identity Component, $ST^0(\text{Jac}(C_{p^2}))$

Since $\text{Jac}(C_{p^2})$ is an abelian variety with CM, we have that

$$ST^0(\text{Jac}(C_{p^2})) \cong \text{Hg}(\text{Jac}(C_{p^2})),$$

where $\text{Hg}(\text{Jac}(C_{p^2}))$ is the **Hodge group** of $\text{Jac}(C_{p^2})$.

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We have that

Proposition [CGHM25]

$\text{Hg}(\text{Jac}(C_{p^2})) \cong \text{U}(1)^{g'}$, where $g' = \phi(p^2)/2$.

Proof idea:

- $\text{Jac}(C_{p^2}) \sim \text{Jac}(C_p) \times X_{p^2}$ and $\text{MT}(\text{Jac}(C_{p^2})) \cong \text{MT}(X_{p^2})$ by [GGL24]
- $\text{Hg}(\text{Jac}(C_{p^2})) \cong \text{Hg}(X_{p^2}) \cong \text{L}(X_{p^2}) \cong \text{U}(1)^{g'}$.

The Identity Component (Cont.)

This result tells us that $\mathrm{Hg}(\mathrm{Jac}(C_{p^2}))$ is **smaller than expected**—since $\mathrm{Jac}(C_{p^2})$ has CM, it'd "normally" be isomorphic to $\mathrm{U}(1)^g$.

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This is reflected by the fact that $\mathrm{Jac}(C_{p^2})$ is **degenerate** (by [Goo24]).

Definition

An abelian variety A is *degenerate* if its Hodge ring

$$\mathcal{B}^*(A) := \sum_{d=0}^{\dim(A)} \mathcal{B}^d(A),$$

where $\mathcal{B}^d(A)$ is the $\mathbb{C}$ -span of the Hodge classes of codimension d on A , contains **exceptional** (Hodge) classes—Hodge classes not generated by classes of codimension $d = 1$ (i.e., divisor classes).

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Since $\mathrm{ST}(\mathrm{Jac}(C_{p^2})) \subseteq \mathrm{USp}(2g)$ and $g - g' = (p - 1)/2$, if we identified an element of $\mathrm{Hg}(\mathrm{Jac}(C_{p^2}))$ with a $U \in \mathrm{U}(1)^g$, **$p - 1$ entries of U are dependent on other entries of U .**

Extracting the Dependencies

Informally, the Hodge ring is made up of the classes that are fixed by the Hodge group ([BL04, Thm. 17.3.3]). So, if $U \in \text{Hg}(\text{Jac}(C_{p^2}))$ (as a matrix) and v is a Hodge class (in the Hodge ring), then

$$U \cdot v = v.$$

This action is how we'll extract the extra relations.

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Identifying the Hodge group with an element from $\text{U}(1)^{g'}$ already incorporates the relations from the divisor classes—it's just

$$\text{diag}(U_1, \overline{U}_1, U_2, \overline{U}_2, \dots, U_{g'}, \overline{U}_{g'}),$$

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We look at the indecomposable Hodge classes—exceptional classes not generated by classes of lower codimension.

Redefining Indecomposable Classes

In [Shi82], Shioda defines a set of tuples that act as an index set for Hodge classes of codimension d :

Definition [CGHM25]

Let m be a positive, odd integer and d be an integer satisfying $1 \leq d \leq \frac{m-1}{2}$. We define the set

$$\mathfrak{B}_m^d := \{\beta = (b_1, b_2, \dots, b_{2d})\}$$

to be the set of tuples of length $2d$ satisfying the following properties:

1. $1 \leq b_1 < b_2 < \dots < b_{2d} \leq m - 1$;
2. $\sum_{i=1}^{2d} b_i \equiv 0 \pmod{m}$;
3. $|t \cdot \beta| = d$ for all $t \in (\mathbb{Z}/m\mathbb{Z})^\times$, where $|t \cdot \beta| = \sum_{i=1}^{2d} \langle tb_i \rangle_m / m$.

Redefining Indecomposable Classes (Cont.)

Namely, he showed that there is a correspondence between tuples in $\mathfrak{B}_m^d$ to Hodge classes:

[Shi82, Thm. 5.2]

Assume m is odd. The Hodge classes on the Jacobian variety $\text{Jac}(C_m)$ have the following description:

$$\mathcal{B}^d(\text{Jac}(C_m)) = \bigoplus_{(b_1, \dots, b_{2d}) \in \mathfrak{B}_m^d} \mathbb{C} \omega_{b_1} \wedge \dots \wedge \omega_{b_{2d}}.$$

So

$$(b_1, b_2, \dots, b_{2d}) \longleftrightarrow \omega_{b_1} \wedge \omega_{b_2} \wedge \dots \wedge \omega_{b_{2d}}.$$

Redefining Indecomposable Classes (Cont.)

So we can frame *exceptional*-ness and *indecomposable*-ness in terms of tuples:

Definition [CGHM25]

We say that a tuple $\beta \in \mathfrak{B}_m^d$ is **exceptional** if it's not entirely made up of pairs b_i, b_j such that $b_i + b_j \equiv 0 \pmod{m}$.

We say that $\beta \in \mathfrak{B}_m^d$ is **indecomposable** if no proper subset (with an even number of elements) of $\{b_1, b_2, \dots, b_{2d}\}$ adds to a multiple of m . Otherwise, we say that β is *decomposable*.

Example: $m = p^2 = 9$, $d = (3 + 1)/2 = 2$

- $(1, 4, 6, 7)$ and $(2, 3, 5, 8)$ are exceptional and indecomposable, but $(1, 2, 7, 8)$ isn't exceptional

Example: $m = p^2 = 25$, $d = 4$

- $(1, 2, 6, 11, 16, 20, 21, 23)$ is exceptional, but not indecomposable

Our Case: $m = p^2$

In the proof of [Shi82, Lemma 5.5], Shioda defined a family of indecomposable tuples of codimension $d = (p + 1)/2$: For $1 \leq i \leq p - 1$, define

$$\beta_i := (i, i + p, i + 2p, \dots, i + (p - 1)p, p(p - i)).$$

(We write β_i to signify the tuple's entries have been permuted to be an element of $\mathfrak{B}_m^d$)

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It turns out, *all* indecomposable tuples (when $m = p^2$) come from β_i . Meaning, the only codimension where indecomposable classes exist is $d = (p + 1)/2$ ([CGHM25, Thm. 3.21]).

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Furthermore, there are exactly $p - 1$ many of these tuples when $m = p^2$ ([CGHM25, Thm. 3.21]).

The Indecomposable Classes Characterized

Using the above correspondence, this means

[CGHM25, Cor. 3.22]

From each indecomposable tuple

$$\beta_i := (i, i + p, i + 2p, \dots, i + (p - 1)p, p(p - i)),$$

the indecomposable Hodge classes of codimension $(p + 1)/2$ are given by

$$\nu_i = \omega_i \wedge \omega_{i+p} \wedge \omega_{i+2p} \wedge \dots \wedge \omega_{i+(p-1)p} \wedge \omega_{p(p-i)},$$

where $1 \leq i \leq p - 1$.

An Adjustment

We'll modify the elements of β_i such that every entry b_j with $j > d = \frac{p+1}{2}$ is written as $b_j - p^2$. This modification will negate elements of the tuple whose value is greater than $\frac{p^2}{2}$. This corresponds to expressing the differential ω_{b_j} as $\bar{\omega}_{p^2-b_j}$.

After modifying the tuples in this way, we obtain pairs of tuples such that each β_i is paired with the corresponding tuple β_{p-i} , where both are negatives of each other.

Ex: $p^2 = 9$

- $\beta_1 = (1, 4, 6, 7) \rightarrow (1, 4, -3, -2) \longleftrightarrow \nu_1 = \omega_1 \wedge \omega_4 \wedge \bar{\omega}_3 \wedge \bar{\omega}_2$
- $\beta_2 = (2, 3, 5, 8) \rightarrow (2, 3, -4, -1) \longleftrightarrow \nu_2 = \omega_2 \wedge \omega_3 \wedge \bar{\omega}_4 \wedge \bar{\omega}_1$

We read off the effect of the Hodge group in every new β_i . So, it's sufficient to just focus on the tuples β_i where $1 \leq i \leq \frac{p-1}{2}$.

New Expression of Indecomposable Classes

From that adjustment of each β_i , we obtain a new expression of the indecomposable Hodge classes

[CGHM25, Cor. 3.26]

Let $1 \leq i \leq \frac{p-1}{2}$. Then

$$\nu_i = \omega_i \wedge \omega_{i+p} \wedge \omega_{i+2p} \wedge \cdots \wedge \omega_{i+p\frac{p-1}{2}} \wedge \overline{\omega}_{p\frac{p-1}{2}-i} \wedge \cdots \wedge \overline{\omega}_{p-i} \wedge \overline{\omega}_{pi}.$$

The Indecomposable Classes Characterized (Cont.)

By the group action $U \cdot \nu_i = \nu_i$, when ν_i is an indecomposable class we have

$$\begin{aligned} U \cdot \nu_i &= U \cdot (\omega_i \wedge \omega_{i+p} \wedge \omega_{i+2p} \wedge \cdots \wedge \omega_{i+p\frac{p-1}{2}} \wedge \bar{\omega}_{p\frac{p-1}{2}-i} \wedge \cdots \wedge \bar{\omega}_{p-i} \wedge \bar{\omega}_{pi}) \\ &= (u_i u_{i+p} u_{i+2p} \cdots u_{i+p\frac{p-1}{2}} \bar{u}_{p\frac{p-1}{2}-i} \cdots \bar{u}_{p-i} \bar{u}_{pi}) \nu_i. \end{aligned}$$

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Since the Hodge group fixes elements from the Hodge ring, we have that

$$u_i u_{i+p} u_{i+2p} \cdots u_{i+p\frac{p-1}{2}} \bar{u}_{p\frac{p-1}{2}-i} \cdots \bar{u}_{p-i} \bar{u}_{pi} = 1.$$

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The largest subscript is $i + p\frac{p-1}{2}$, so isolating it gives

$$u_{i+p\frac{p-1}{2}} = \bar{u}_i \bar{u}_{i+p} \bar{u}_{i+2p} \cdots \bar{u}_{i+p\frac{p-3}{2}} u_{p\frac{p-1}{2}-i} \cdots u_{p-i} u_{pi}$$

and

$$\bar{u}_{i+p\frac{p-1}{2}} = u_i u_{i+p} u_{i+2p} \cdots u_{i+p\frac{p-3}{2}} \bar{u}_{p\frac{p-1}{2}-i} \cdots \bar{u}_{p-i} \bar{u}_{pi}.$$

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These are *exactly* the missing relations!

The Identity Component, Revisited

By the previous slide, we can now express $ST^0(\text{Jac}(C_{p^2}))$:

[CGHM25, Prop. 4.1]

The identity component of the Sato-Tate group of $\text{Jac}(C_{p^2})$ is isomorphic to $U(1)^{g'}$. We can identify elements of the identity component with matrices $U = \text{diag}(U_1, U_2, \dots, U_g)$ in $U(1)^g$ where

$$U_{i+p\frac{p-1}{2}} = \overline{U}_i \overline{U}_{i+p} \overline{U}_{i+2p} \cdots \overline{U}_{i+p\frac{p-3}{2}} U_{p\frac{p-1}{2}-i} \cdots U_{p-i} U_{pi}$$

for $1 \leq i \leq \frac{p-1}{2}$.

Example of Identity Component: C_{25}

Let $p = 5$. The genus of C_{25} is $g = (25 - 1)/2 = 12$. The only indecomposable tuples are

$(1, 6, 11, 16, 20, 21)$, $(2, 7, 12, 15, 17, 22)$, $(3, 8, 10, 13, 18, 23)$, $(4, 5, 9, 14, 19, 24)$
and they're all of the form β_i with $1 \leq i \leq 4$.

We select the first two tuples and adjust each entry of the tuple whose value is greater than $p^2/2 = 12.5$ to obtain

$$\beta_1 = (1, 6, 11, -9, -5, -4) \quad \beta_2 = (2, 7, 12, -10, -8, -3).$$

These correspond to the Hodge classes

$$\nu_1 = \omega_1 \wedge \omega_6 \wedge \omega_{11} \wedge \bar{\omega}_9 \wedge \bar{\omega}_5 \wedge \bar{\omega}_4 \quad \text{and} \quad \nu_2 = \omega_2 \wedge \omega_7 \wedge \omega_{12} \wedge \bar{\omega}_{10} \wedge \bar{\omega}_8 \wedge \bar{\omega}_3.$$

Let $U \in U(1)^g$. We get the following relations among entries of U

$$u_{11} = \bar{u}_1 u_4 u_5 \bar{u}_6 u_9 \quad \text{and} \quad u_{12} = \bar{u}_2 u_3 \bar{u}_7 u_8 u_{10},$$

giving us the identity component

$$U = \text{diag}(U_1, U_2, \dots, U_{10}, \bar{U}_1 U_4 U_5 \bar{U}_6 U_9, \bar{U}_2 U_3 \bar{U}_7 U_8 U_{10}).$$

The Sato-Tate Group of $\text{Jac}(C_{p^2})$

[CGHM25, Thm. 4.6]

Let $g = \frac{p^2-1}{2}$ be the genus of the curve C_{p^2} and let $g' = \frac{p(p-1)}{2}$. The Sato-Tate group of $\text{Jac}(C_{p^2})$, up to isomorphism in $\text{USp}(2g)$, is given by

$$\text{ST}(\text{Jac}(C_{p^2})) \simeq \langle \text{U}(1)^{g'}, \gamma \rangle,$$

where the embedding of $\text{U}(1)^{g'}$ in $\text{USp}(2g)$ is described in Slide 21.

(a_1) Moment Statistics of C_{25}

The numerical moments coming from the a_1 coefficient of the normalized L-polynomial were computed up to primes $p < 2^{25}$

	M_2	M_4	M_6	M_8
a_1	2.009	90.848	9452.007	1438061.241
μ_1	2	90	9344	1419866

Table 1: Table of a_1 - and μ_1 -moments for $C_{25} : y^2 = x^{25} - 1$ (with $p < 2^{25}$).

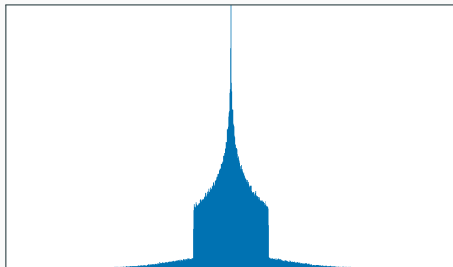
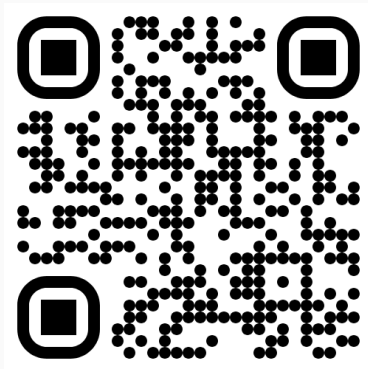


Figure 1: Histogram of the a_1 -coefficients for $C_{25} : y^2 = x^{25} - 1$.

Thank you!

Read our paper!



or search [Degeneracy and Sato-Tate Groups of \$y^2 = x^{p^2} - 1\$](#)

Bonus: Sato-Tate Group Definition

For an abelian variety A of dimension g over a field F and prime ℓ , the Galois action on the Tate module is given by an ℓ -adic representation

$$\rho_{A,\ell} : \mathrm{Gal}(\bar{F}/F) \rightarrow \mathrm{Aut}(V_\ell) \cong \mathrm{GL}_{2g, \mathbb{Q}_\ell},$$

where V_ℓ is the rational Tate module.

The ℓ -adic monodromy group of A , denoted as $G_{A,\ell}$, is the Zariski closure of the image of this map over $\mathrm{GL}_{2g, \mathbb{Q}_\ell}$. Additionally, let $G_{A,\ell}^1 := G_{A,\ell} \cap \mathrm{Sp}_{2g, \mathbb{Q}_\ell}$.

Definition [Goo24, Sec. 2.4]

The **Sato-Tate group** of A , denoted as $\mathrm{ST}(A)$, is a maximal compact Lie subgroup of $G_{A,\ell}^1 \otimes_{\mathbb{Q}_\ell} \mathbb{C}$ contained in $\mathrm{USp}(2g)$.

Bonus: Moment Statistics

Moment statistics from the $ST(\text{Jac}(C_{p^2}))$ are called **theoretical** moments, whereas those from the normalized L-polynomials are called **numerical** moments.

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By the isomorphism of $\text{ST}(\text{Jac}(C_{p^2}))$, we can compute moments by working with $\langle \text{U}(1)^{g'}, \gamma \rangle$ instead.

Bonus: Moment Statistics (Cont. + Some Background)

For the unitary group $U(1)$, the trace map tr on a random element $U \in U(1)$ is given by $z := \text{tr}(U) = u + \bar{u} = 2 \cos(\theta)$, where $u = e^{i\theta}$. Then $dz = -2 \sin(\theta) d\theta$ and

$$\mu_{U(1)} = \frac{1}{2\pi} \frac{dz}{\sqrt{4 - z^2}} = \frac{1}{2\pi} d\theta$$

gives a uniform measure of $U(1)$ on the eigenangle $\theta \in [-\pi, \pi]$ (see [Sut19, Section 2]). The n^{th} moment $M_n[\mu]$ is the expected value of $\phi_n : z \mapsto z^n$ with respect to μ , computed as

$$M_n[\mu] = \int_I z^n \mu(z),$$

where $I = [-2, 2]$.

Bonus: Moment Statistics (Cont.)

Let U be a random matrix in $ST^0(\text{Jac}(C_{p^2}))$ and γ be the component group generator. Denote

$$g_i^k$$

to be the coefficient of T^i in the characteristic polynomial of $U\gamma^k$ (where $0 \leq k \leq \phi(p^2)$).

Note: Frob_p is defined up to conjugacy, so we can think of $\rho_{A,\ell}(\text{Frob}_p)$ —a matrix—as representing a conjugacy class. Thus, working with $ST(A)$ charpolys means inherently working with its conjugacy classes.

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The n th moment $M_n[\mu_i^k]$ is then the expected value of $(g_i^k)^n$, and we compute this by integrating against the Haar measure. Once done, we obtain moment statistics for the entire Sato-Tate group by taking the average of the moments for $U\gamma^k$.

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Bonus: Example of Moment Statistics: C_{25}

Let $p = 5$ ($g = 12$). We first compute the characteristic polynomial of each $U\gamma^k$, where $0 \leq k \leq \phi(25) = 20$.

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Even more surprising, g_1^0 ($k = 0$) has the *largest* number of terms. Naturally, this creates the most complicated integral...

When $k = 0$, $M_n[\mu_1^0]$ is equal to the value of the following integral

$$\frac{2^n}{(2\pi)^{10}} \int_0^{2\pi} \cdots \int_0^{2\pi} (\cos(\theta_1) + \cdots + \cos(\theta_{10}) \\ + \cos(-\theta_1 + \theta_4 + \theta_5 - \theta_6 + \theta_9) + \cos(-\theta_2 + \theta_3 - \theta_7 + \theta_8 + \theta_{10}))^n d\theta_1 \cdots d\theta_{10}.$$

We can see degeneracy manifesting in the last two terms, since we're taking the n th moment of just U here.

Bonus: Example of Moment Statistics: C_{25} (Cont.)

To compute $M_n[\mu_1^k]$ for $k = 4, 8, 12, 16$, we integrate

$$\frac{(\pm 2)^n}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} (\cos(\theta_5) + \cos(\theta_{10}))^n d\theta_5 d\theta_{10},$$

where the numerator of the coefficient is 2^n when $k = 4, 12$ and $(-2)^n$ when $k = 8, 16$.

Bonus: Example of Moment Statistics: C_{25} (Cont.)

To compute $M_n[\mu_1^k]$ for $k = 4, 8, 12, 16$, we integrate

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where the numerator of the coefficient is 2^n when $k = 4, 12$ and $(-2)^n$ when $k = 8, 16$.

We then derive the full moment statistics $M_n[\mu_1]$ of the full Sato-Tate group by averaging over the size of the group (i.e., compute up to some moment for each restriction, then divide said moments by the size of the group).

Bonus: L-Polynomials

For primes p of good reduction for C , the [zeta function](#) of C is

$$Z(C/\mathbb{F}_p, T) := \exp \left(\sum_{k=1}^{\infty} \frac{\#C(\mathbb{F}_{p^k}) T^k}{k} \right) = \frac{L_p(C, T)}{(1-T)(1-pT)}.$$

Define the normalized L -polynomial as

$$\begin{aligned} \bar{L}_p(C, T) &:= L_p(C, T/\sqrt{p}) \\ &= T^{2g} + a_1 T^{2g-1} + a_2 T^{2g-2} + \cdots + a_2 T^2 + a_1 T + 1, \end{aligned}$$

where $a_i \in \left[-\binom{2g}{i}, \binom{2g}{i} \right]$ and g denotes the genus of C .

The coefficients of $\bar{L}_p(C, T)$ contain important arithmetic information about C

- The a_1 coefficient is the *trace of Frobenius*:

$$a_1 = p + 1 - \#C(\mathbb{F}_p).$$

Bonus: Cyclicity of $(\mathbb{Z}/p^2\mathbb{Z})^\times$

- The map

$$f: \mathbb{Z}/p^2\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$$

is a surjective ring homomorphism which restricts to a surjective group homomorphism

$$g: (\mathbb{Z}/p^2\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times.$$

- From the group homomorphism,

$$(\mathbb{Z}/p^2\mathbb{Z})^\times \cong \ker(g) \times (\mathbb{Z}/p\mathbb{Z})^\times,$$

where $\ker(g)$ and $(\mathbb{Z}/p\mathbb{Z})^\times$ are finite cyclic groups of coprime orders.

- Product of two cyclic groups of coprime orders is itself a cyclic group, so $(\mathbb{Z}/p^2\mathbb{Z})^\times$ is a cyclic group.

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